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Before Starting...

Go to:

github.com/ffusillo/daisy-networks

Network analysis: indicators, mapping and empirics with innovation data

DAISY International Summer School · hands-on session

Fabrizio Fusillo · University of Turin

Taranto, 7-11 September 2026



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01 Why networks for innovation?

Three reasons - and one methodological consequence

What is a network?

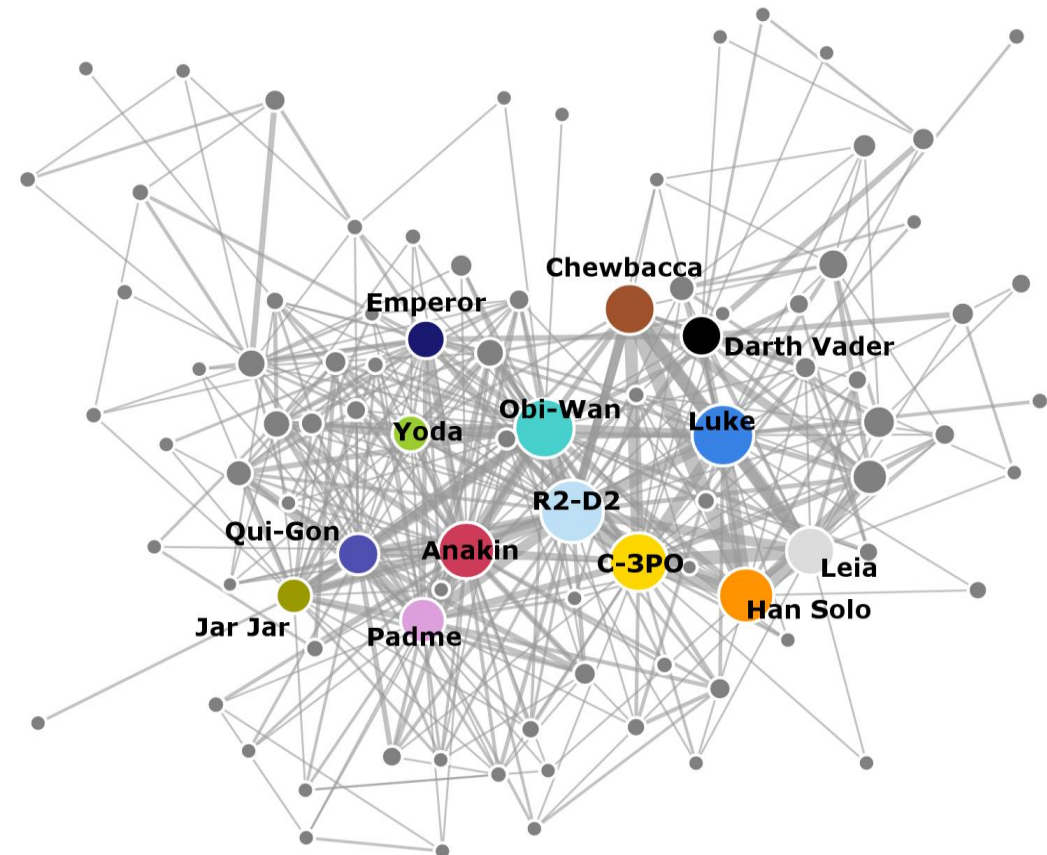


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What is a network?

- A set of entities (nodes) that have some sort of relationship between them (ties)
- Network is its representation



Networks are everywhere

- Reality is complicated
- Complex systems
 - difficult to derive their collective behaviour from a knowledge of the system's components
- Behind each complex system there is an intricate network that encodes the interactions between components



Innovation is a collective process

- Innovation is a **collective (interactive) process**

Outcome of collective efforts of heterogeneous actors...
that interact and collaborate
recombining existing and novel pieces of knowledge

- Firms rely increasingly on external sources of knowledge, capabilities, and complementary assets
- **Knowledge flows** between actors become a fundamental engine for innovation
- Networks are the locus of innovation
- Observable as: R&D collaboration, co-invention, co-publication, alliances, EU consortia

Three orders of relationship



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- ORIGIN - why collaborate? Who collaborate? With whom? How?
- STRUCTURE - what the network looks like: density, clustering, components, small worlds, fat tails
- EFFECT – is the position within the network associated to «performances»?

→ **Decide which order you are studying before you choose a measure**

Why this matters for sustainability

- Green/circular/twin technologies recombine often distant domains
energy x digital, mobility x materials, chemistry x biology
 - They lean more on public research and on policy-created collaboration than the average technology
 - Direction of change is also network phenomenon: who is connected decides what gets recombined next
 - So transitions are studied as changes in structure, not only as changes in volumes
- **Green technologies are complex and sophisticated – integrate heterogeneous knowledge sources**



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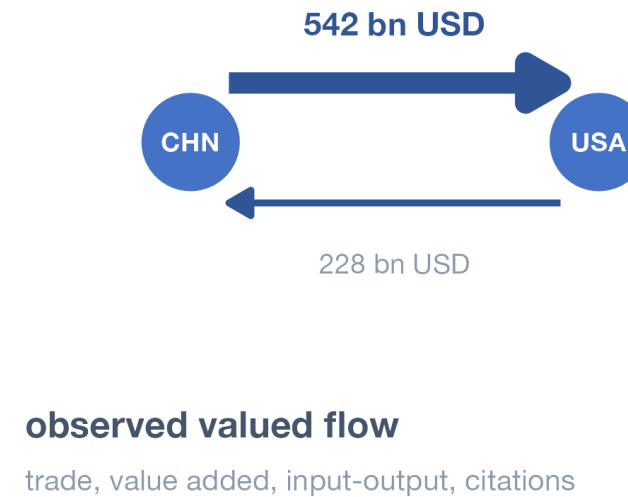
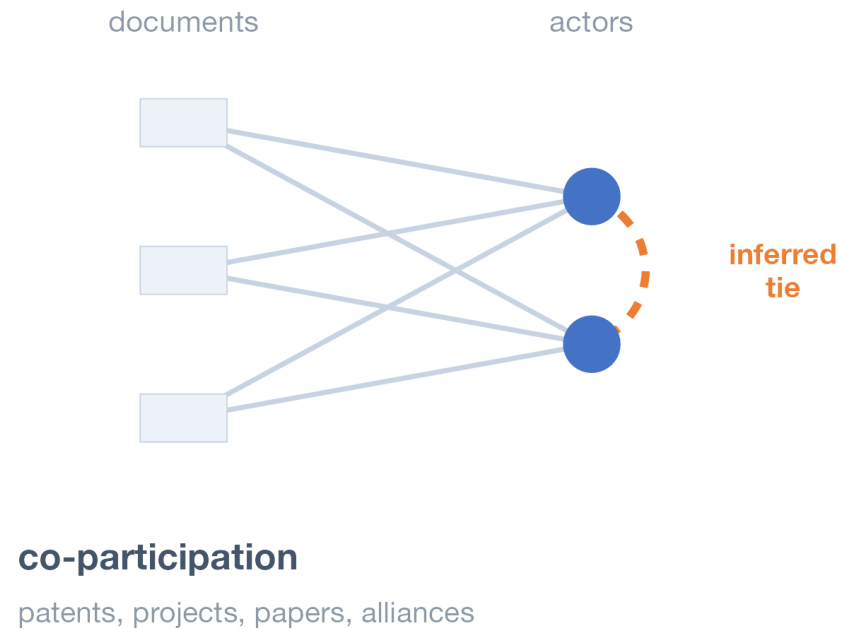
02 Where does the data come from?

Two families of relational data

Two families of relational data



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Left: the tie is inferred from co-participation - a modelling choice. Right: the tie is given, with a direction and a magnitude, and the network is nearly complete, so the method becomes the filter.

From two modes to one

	a1	a2	a3	a4	a5	a6
e1						
e2						
e3						
e4						
e5						

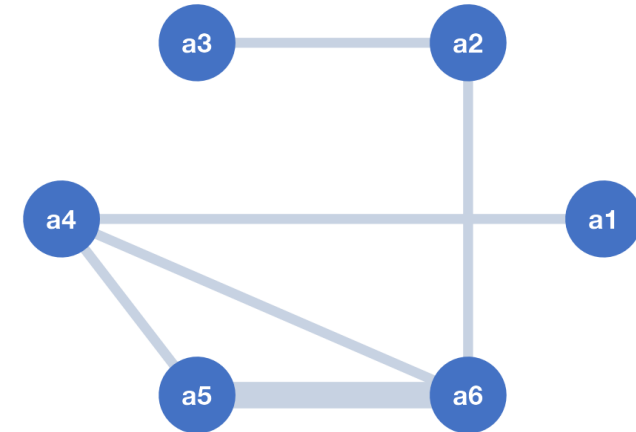
B : events x actors

$$A = B'B$$



one sparse crossproduct

weight = shared events

**A : actors x actors**

Every event becomes a clique. The same operation works at any level - inventors, organisations, regions, technologies, products

Five sources, and what each one is for

Source	The tie is	Levels available	Access
OECD REGPAT / PATSTAT	co-invention, co-application, citations	inventor, firm, region, technology	REGPAT free on request
CORDIS (Horizon Europe)	co-participation in a funded project	organisation, region, country, topic	open, CC-BY
OpenAlex	co-authorship, citations	author, institution, country	open, CC0, no key
OECD TiVA	value added embodied in final demand	country, country x industry	open, SDMX API
CEPII BACI	gross exports by product	country, country x product	free, cite CEPII



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03 Measures

What to compute, and what it actually means

Three levels, three questions



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NODE

degree · strength · betweenness · eigenvector · constraint · brokerage roles

who matters, and in which sense?

NETWORK

density · components · clustering · assortativity · communities · centralisation

what does the whole thing look like?

INDICATOR

relatedness · variety · coherence · complexity · relatedness density

what number goes into the model?

The level you choose is not a technical detail, is a modelling choice

Node Centrality

- Focus on **individual nodes** and their individual property:
 - its *position* in a network
- Measures the **structural importance** of a node in a network
 - How do you measure importance?
 - Power? Information flows? Overall network stability?
- Can be interpreted in **several ways** and measured with a variety of indicators
- Empirically, it is often employed as an independent variable to predict nodes outcomes

Network level: what the whole thing looks like

- Size: describe the dimension of the network
- Cohesion: level of connectedness in a network
- Centralization: is the network dominated by a single/few nodes? Core-periphery

Brokerage: four different questions

THE FOUR IDEAS

- BETWEENNESS - global flow control

THE MEASURES

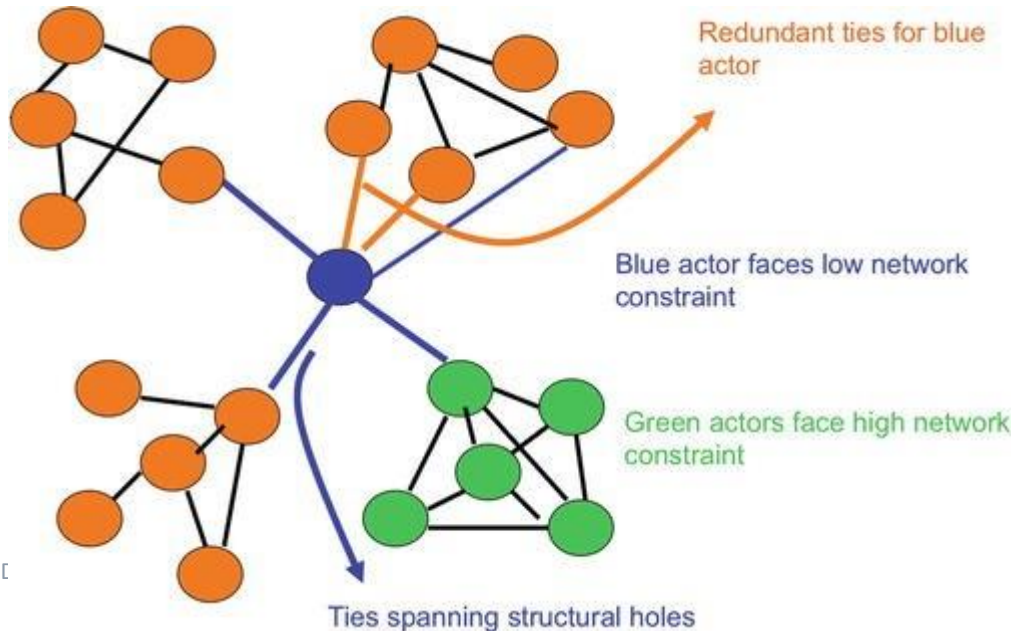
It counts the number of times a node is “in-between” the shortest paths between all the other nodes

→ **Four measures, four questions - and they rank different actors first**

Brokerage: four different questions

THE FOUR IDEAS

- BETWEENNESS - global flow control
- STRUCTURAL HOLES - are my contacts disconnected from each other?



THE MEASURES

Effective Size - measures the non-redundant portion of an ego's network, representing the number of independent contacts the ego maintains

Constraint - Constraint measures the extent to which an ego's connections are invested in people who are interconnected with one another

Brokerage: four different questions

THE FOUR IDEAS

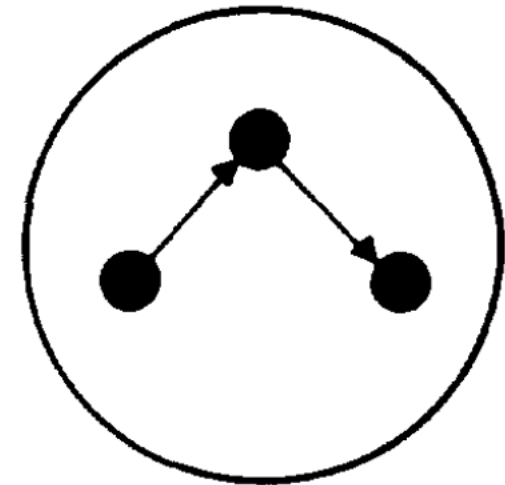
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- STRUCTURAL HOLES - are my contacts disconnected from each other?
- GOULD-FERNANDEZ - which gap: coordinator, gatekeeper, representative, consultant, liaison

→ **Four measures, four questions - and they rank different actors first**

THE MEASURES

Local broker (coordinator)

All the three actors belong to the same group



Brokerage: four different questions

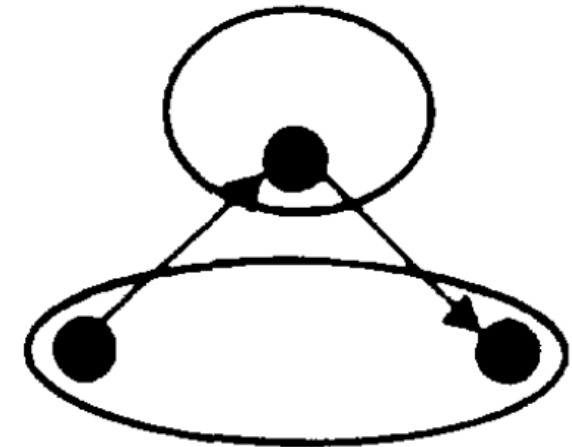
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THE MEASURES

Itinerant broker

Two actors belong to the same group, the mediator to a different group



→ Four measures, four questions - and they rank different actors first

Brokerage: four different questions

THE FOUR IDEAS

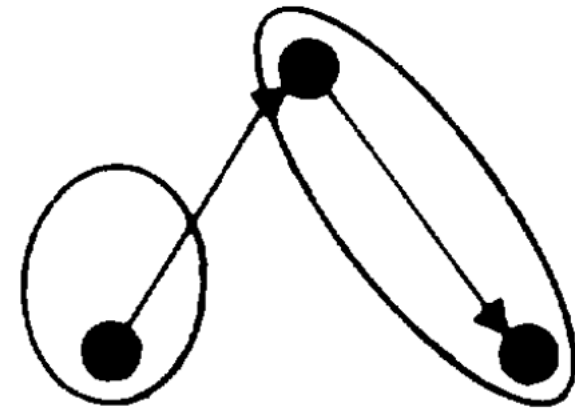
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THE MEASURES

Gatekeeper broker

The mediator and one of the mediated belong to the same group



Brokerage: four different questions

THE FOUR IDEAS

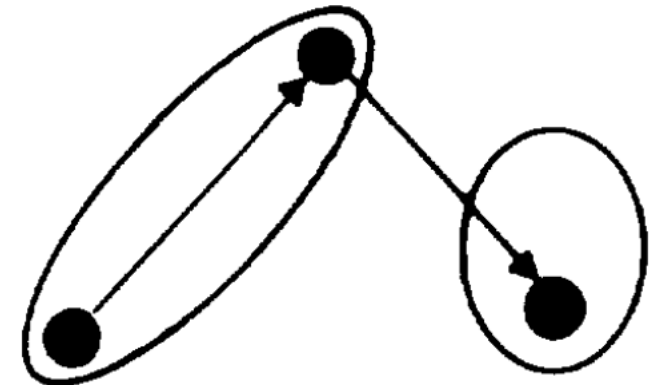
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THE MEASURES

Representative broker

The mediator and one of the mediated belong to the same group



Brokerage: four different questions

THE FOUR IDEAS

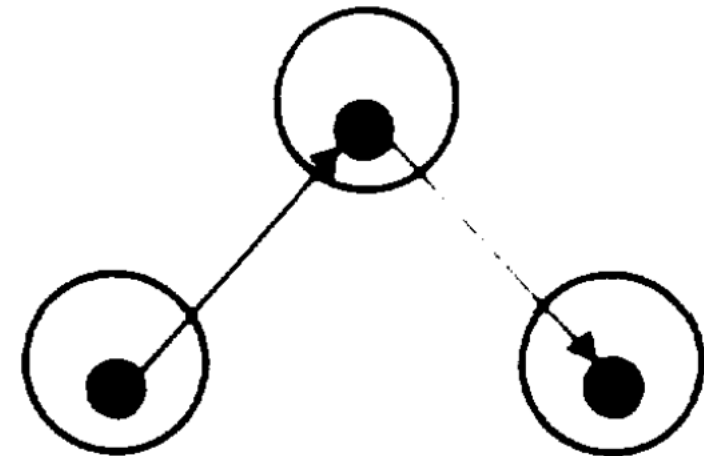
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- STRUCTURAL HOLES - are my contacts disconnected from each other?
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THE MEASURES

Liaison broker

All the three actors belongs to different groups



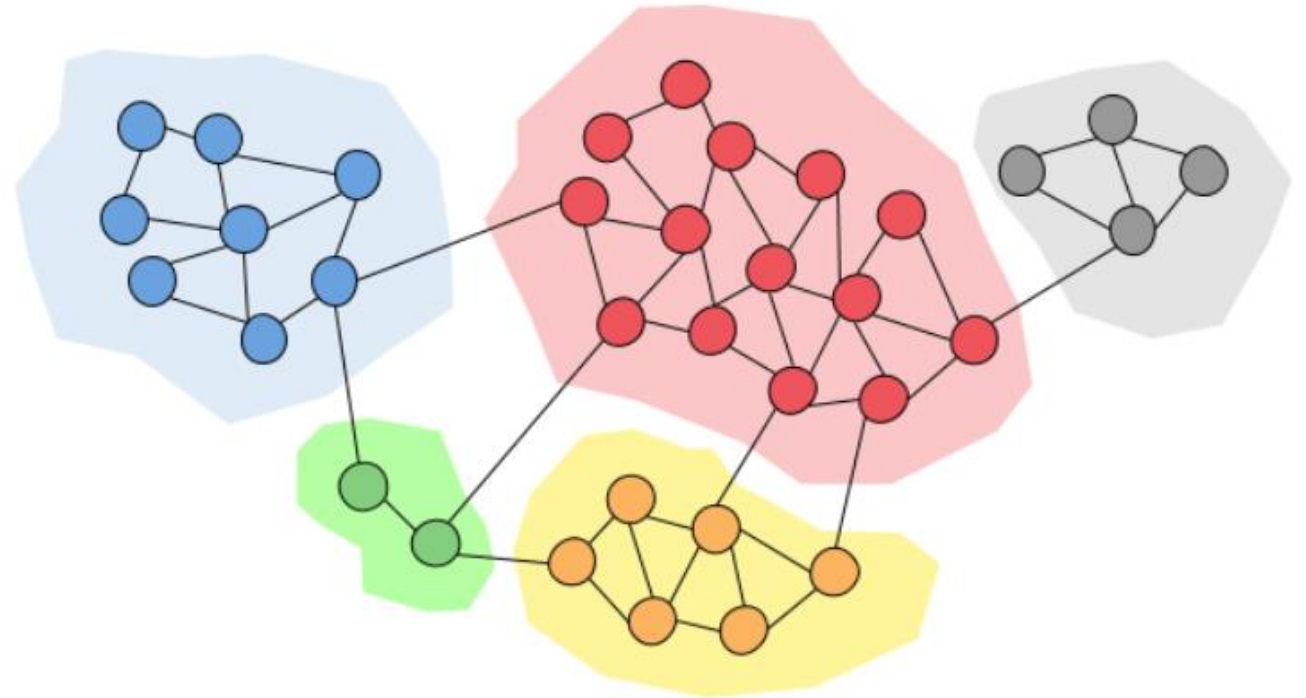
Communities detection: a partition is a claim



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THE CHOICES YOU MAKE

- Several different algorithms: Louvain / Leiden (modularity), Infomap (flow), walktrap
- Yielding different results



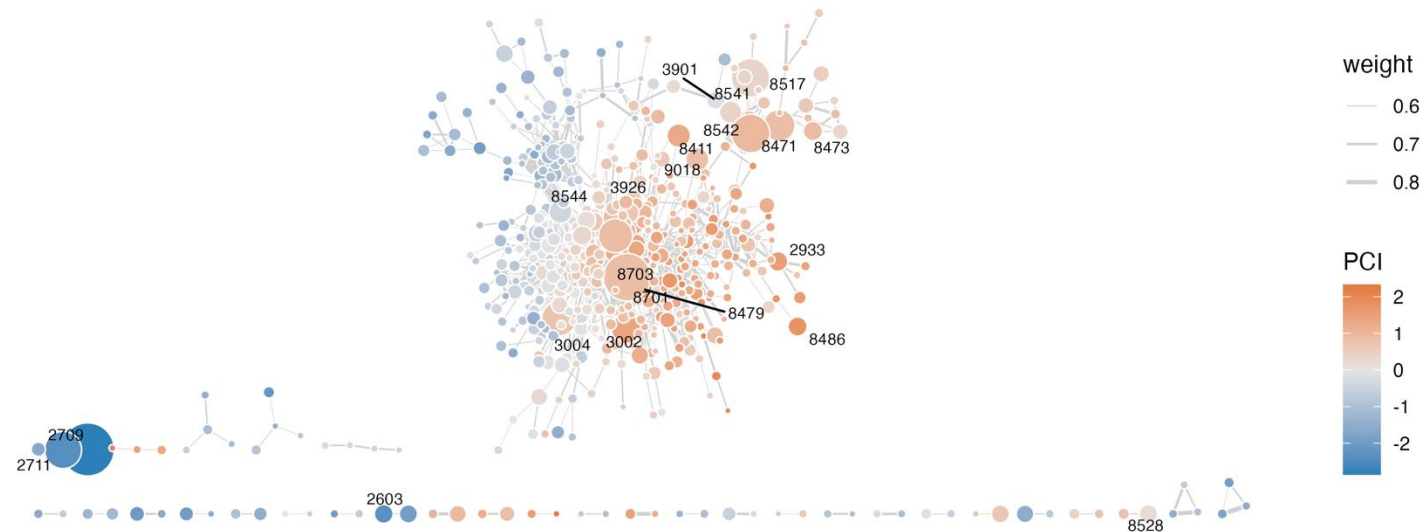
Networks as measurement devices

- Sometimes the network is not the object of interest but the instrument
- RELATEDNESS: co-classification within documents, or co-specialisation across places
- KNOWLEDGE / PRODUCT SPACE: the map those proximities draw
- VARIETY (entropy; related vs unrelated) and COHERENCE (average relatedness of a portfolio)
- COMPLEXITY (ECI/KCI, PCI/TCI): diversity and ubiquity, reflected on each other
- RELATEDNESS DENSITY: how close a new activity is to what you do - it predicts entry

→ **The network you do not need to draw is often the one that ends up in the regression**



The same construction: the product space



BACI 2023, HS4 products, proximity from co-specialisation across countries; colour = complexity (PCI).

202

technologies (CPC4, patents)

83

topics (euroSciVoc, projects)

582

products (HS4, exports)



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